

NOTES & COMMENTS

Liposuction for axillary hyperhidrosis: Reconciling trial results and expert opinion

To the Editor: We would like to thank Dr Coleman¹ for his insightful comments regarding our article and for sharing his expertise in the use of liposuction for treatment of axillary hyperhidrosis. As Dr Coleman notes, and as we have discussed elsewhere,^{2,3} one of the major challenges of comparative effectiveness studies of procedural interventions is so-called “dose-finding.” That is, selection of the appropriate settings, intensity, and precise surgical technique for each treatment arm is needed to ensure that a fair, apples-to-apples comparison of the procedures is performed. The more complex the interventions, the more numerous the variables that need to be managed, and the harder this is to accomplish. A similar challenge occurs when techniques are not standardized, or when they are highly operator-dependent.

Dr Coleman notes that he performs a more aggressive procedure than we selected, and we could have certainly included one such in our study. But then another commentator could have asserted that our very aggressive liposuction procedure was outside the norm, and biased our results in favor of liposuction. Yet another consideration is the institutional review board process, including informed consent and subject risk reduction. Botulinum toxin injections pose extremely low risks, and while liposuction is also very safe, it is an invasive procedure with a slightly elevated risk profile, including the possibility of postoperative edema and erythema, scars, and hyperpigmentation. This is another reason we chose to use a slightly less aggressive procedure.

We also agree with Dr Coleman’s observation that the duration of our study precluded long-term outcome assessment. This is a limitation of many studies, including ours, and is a particular issue in studies of cosmetic or lifestyle interventions,^{2,3} in which patients are often reluctant to return for additional follow-up visits years later.

That being said, we believe that randomized controlled trials do offer information that is supplementary to expert opinion. Assuming they are reasonably well designed, randomized controlled trials do mitigate biases inherent to observational studies or uncontrolled trials. Therefore, large effect sizes that are sometimes seen in observational studies often shrink when randomized controlled trials are performed.

Additional strengths of our study include the split-body design, which is useful for minimizing unknown confounders, and rare in procedural studies such as ours that involve an invasive treatment. Also, while no study is entirely free of investigator biases or conflicts of interest, we tried to minimize these by avoiding any direct or indirect corporate sponsorship.

We are grateful to Dr Coleman for sharing his expertise as one of the foremost international experts on the procedures we studied, and on dermatologic surgery study design. As Dr Coleman implies, treatment decisions should be based on the totality of the evidence, not the results of any one study or trial. Like Dr Coleman, we look forward to the results of additional studies, including additional comparative effectiveness trials examining microwave technology.

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